

Prototype embeddings for immune receptors: geometry, statistics, and the TCREMP coordinate system

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Appendix T — TCREMP embedding geometry

Abstract

TCREMP [1] embeds an immune receptor as the vector of its alignment distances to a fixed set of *prototype* receptors sampled from the V(D)J recombination model [2, 3]. This appendix gives the mathematics behind that construction. We first develop the general theory of *prototype* (landmark) embeddings of a metric space — their Lipschitz and isometry properties, the negative-type condition under which the alignment dissimilarity is an honest squared Hilbert distance [4], the energy functional that makes embedding-space Euclidean distance track pairwise alignment distance, and a landmark sample-complexity bound (§T.1–§T.2). We then specialise to immune receptors: taking the prototype measure to be the generation distribution P_{gen} turns each coordinate into a Monte-Carlo sketch of a receptor’s relationship to the generative repertoire, which explains why the choice of prototype source scarcely matters and yields the observed Gamma / extreme-value distribution laws (§T.3). The three-component structure of the TCR embedding and paired-chain concatenation are read off as a product metric, with the germline blocks provably low-rank (§T.4); somatic hypermutation is a bounded perturbation that explains why IGH is the hardest chain (§T.5); and the pushforward of P_{gen} onto embedding space is the null model for density-based background subtraction (§T.6). Every quantitative claim is reproduced in-repo (§T.7, Table 1) by `mir/bench/theory.py`.

T.1. FOUNDATIONS OF PROTOTYPE EMBEDDINGS

SETUP. Let (X, d) be a (semi)metric space of receptor sequences: d is symmetric, $d(x, x) = 0$, and — as we verify for the alignment dissimilarity in §T.2 — non-negative. Fix an ordered set of *prototypes* $P = \{p_1, \dots, p_K\} \subset X$. The **prototype embedding** is the landmark map

$$\varphi : X \rightarrow \mathbb{R}^K, \quad \varphi(x) = (d(x, p_1), d(x, p_2), \dots, d(x, p_K)), \quad (1)$$

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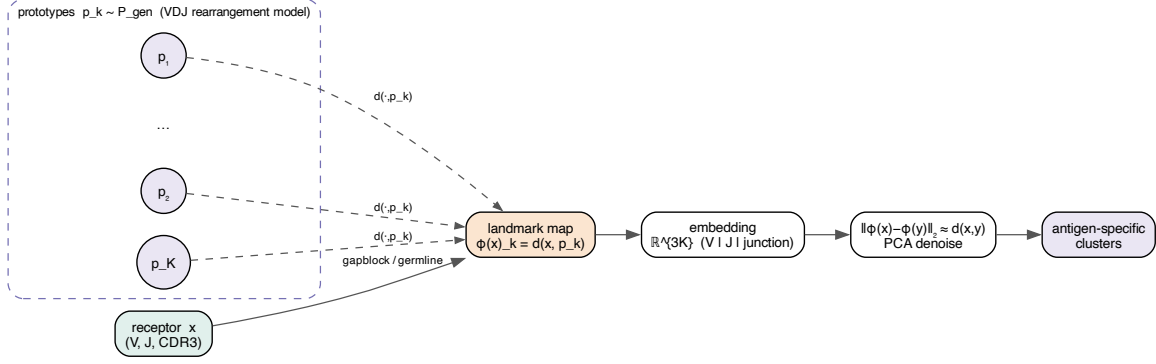


Figure 1: The TCREMP prototype-embedding pipeline. A receptor x is mapped to the vector of its alignment distances to prototypes p_k sampled from the recombination model P_{gen} ; Euclidean distance in the resulting $\mathbb{R}^{3K}$ coordinate system approximates pairwise alignment distance (Prop. 1–3), and specificity groups become compact clusters.

and the induced **embedding distance** is $D(x, y) = \|\varphi(x) - \varphi(y)\|_2$. Figure 1 shows the pipeline. In TCREMP the prototypes are drawn from P_{gen} , d is a BLOSUM alignment dissimilarity, and φ carries three coordinate blocks per prototype (§T.4); the foundational results below hold for any (X, d, P) .

NON-EXPANSIVENESS. The single structural fact from which everything else follows is that each coordinate of φ is 1-Lipschitz.

PROPOSITION 1 (Lipschitz coordinates and the upper bound). *For every prototype p and all $x, y \in X$, $|d(x, p) - d(y, p)| \leq d(x, y)$. Consequently*

$$\|\varphi(x) - \varphi(y)\|_\infty \leq d(x, y) \quad \text{and} \quad D(x, y) = \|\varphi(x) - \varphi(y)\|_2 \leq \sqrt{K} d(x, y). \quad (2)$$

Proof. The triangle inequality gives $d(x, p) \leq d(x, y) + d(y, p)$ and $d(y, p) \leq d(x, y) + d(x, p)$, so $|d(x, p) - d(y, p)| \leq d(x, y)$. Taking the maximum over $p \in P$ gives the ℓ^∞ bound; summing the squares of K terms each $\leq d(x, y)^2$ and taking the root gives the ℓ^2 bound. $\square$ $\square$

Proposition 1 is the rigorous half of the headline “embedding distance $\leq$ alignment distance” claim: no pair is ever *farther* apart in embedding space (up to $\sqrt{K}$) than its alignment distance. The reverse — that close-in-embedding implies close-in-alignment — is not automatic and is where the prototypes must do work.

EXACTNESS IN THE LIMIT. With enough prototypes the ℓ^∞ bound is tight.

PROPOSITION 2 (Kuratowski isometry). *If $P = X$ then $\|\varphi(x) - \varphi(y)\|_\infty = d(x, y)$ for all x, y ; that is, φ is an isometric embedding of (X, d) into $(\ell^\infty, \|\cdot\|_\infty)$.*

Proof. The upper bound is Prop. 1. For the lower bound take the prototype $p = y$: then $|d(x, y) - d(y, y)| = d(x, y)$, so the supremum over prototypes is at least $d(x, y)$. $\square$ $\square$

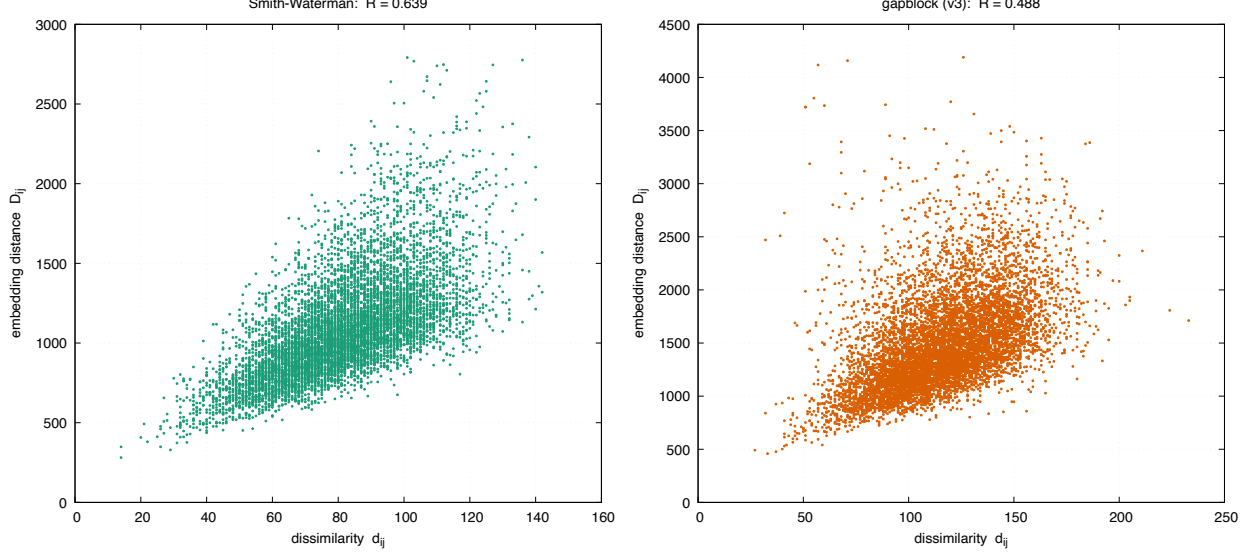


Figure 2: Embedding-space Euclidean distance D_{ij} against pairwise dissimilarity d_{ij} for $n = 2500$ CDR3 β used as their own prototypes (Prop. 3; reproduces suppl. S2). Left: Smith–Waterman d ; right: the gapblock metric shipped in v3. Generated by `mir/bench/theory.py::s2_dissimilarity_distance_correlation`.

The construction of Eq. (1) is thus the Kuratowski/Fréchet embedding, and the supplementary experiments that use the sequences themselves as prototypes (§S1–S2 of [1]) are exactly this isometric regime, read out in ℓ^2 rather than ℓ^∞ . Passing from ℓ^∞ to ℓ^2 costs the $\sqrt{K}$ factor of Eq. (2) but makes the coordinates amenable to PCA and Euclidean clustering.

WHY ℓ^2 DISTANCE TRACKS ALIGNMENT DISTANCE. Fix a prototype-sampling measure μ on X (in TCREMP, $\mu = P_{\text{gen}}$) and draw $p_1, \dots, p_K$ i.i.d. from μ .

PROPOSITION 3 (Energy functional). *Define the μ -energy dissimilarity $\bar{D}_\mu^2(x, y) := \mathbb{E}_{p \sim \mu}[(d(x, p) - d(y, p))^2]$. Then $\frac{1}{K}D(x, y)^2 \rightarrow \bar{D}_\mu^2(x, y)$ almost surely as $K \rightarrow \infty$, and $\bar{D}_\mu^2$ is a μ -dependent (semi)metric with $0 \leq \bar{D}_\mu^2(x, y) \leq d(x, y)^2$.*

Proof. $\frac{1}{K}D^2 = \frac{1}{K} \sum_k (d(x, p_k) - d(y, p_k))^2$ is an empirical mean of i.i.d. terms bounded by $d(x, y)^2$ (Prop. 1); the strong law gives the limit and the bound, and symmetry plus the triangle inequality for $\|\cdot\|_2$ make $\bar{D}_\mu$ a semimetric. □ □

Proposition 3 says D estimates a fixed monotone-in- d functional $\bar{D}_\mu$: two receptors that are close under d have similar distance profiles to the whole of μ and hence small $\bar{D}_\mu$, while dissimilar receptors have systematically different profiles. This is the mechanism behind supplementary Fig. S2 — the empirical Pearson correlation between D_{ij} and d_{ij} (Fig. 2, $R = 0.64$ under Smith–Waterman, 0.49 under the gapblock metric of §T.2) — and behind S2c/d, where D_{ik} concordates with means of d_{ik} over the prototype set: those means are Monte-Carlo estimates of $\bar{D}_\mu$.

HOW MANY PROTOTYPES. The $\sqrt{K}$ gap of Eq. (2) is worst-case; for the *random* landmark map the relative fluctuation of $\frac{1}{K}D^2$ around $\bar{D}_\mu^2$ is $O(K^{-1/2})$ by Prop. 3, so a Johnson–Lindenstrauss-style argument [5] on the bounded coordinates gives the following.

PROPOSITION 4 (Landmark sample complexity). *Let $S \subset X$ with $|S| = n$. For $K = O(\varepsilon^{-2} \log n)$ i.i.d. prototypes, with high probability every pair $x, y \in S$ satisfies $|\frac{1}{K}D(x, y)^2 - \bar{D}_\mu^2(x, y)| \leq \varepsilon d(x, y)^2$ simultaneously.*

Sketch. Each coordinate difference $(d(x, p) - d(y, p))^2 \in [0, d(x, y)^2]$ is bounded; Hoeffding gives a $\exp(-2K\varepsilon^2)$ tail for the deviation of the K -average from $\bar{D}_\mu^2$ per pair, and a union bound over the $\binom{n}{2}$ pairs closes with $K = \Theta(\varepsilon^{-2} \log n)$. $\square$ $\square$

The logarithmic dependence on n is the theoretical counterpart of supplementary Fig. S4, where the UMAP of an antigen-specific set stabilises once $K \gtrsim 100$ prototypes are used: the embedding geometry saturates at a prototype count that grows only logarithmically with the data. Prototype embeddings are the *empirical* / Nyström feature map of the alignment kernel, and Landmark MDS [6, 7] and the dissimilarity representation [8] are the classical instances; the PCA step of §T.4 is Landmark MDS on φ .

T.2. THE ALIGNMENT DISSIMILARITY AS A METRIC

THE GRAM CONSTRUCTION. TCREMP scores two sequences by an alignment similarity s and converts it to a dissimilarity by

$$d(a, b) = s(a, a) + s(b, b) - 2s(a, b). \quad (3)$$

This is exactly the squared distance $\|\psi(a) - \psi(b)\|^2$ induced by a feature map ψ whenever s is an inner product, $s(a, b) = \langle \psi(a), \psi(b) \rangle$. The `v3` implementation (`mir/distances/junction.py`) reads d off directly: `seqtree's` `blosum62()` is already the Gram transform of Eq. (3), and `gapblock.score_matrix` returns d as a genuine metric (zero diagonal, symmetric, non-negative), selecting the best of contiguous gap-block placements at offsets (3, 4, −4, −3).

DEFINITION 1. A symmetric d with zero diagonal is of *negative type* (equivalently, $-d$ is conditionally positive definite) if $\sum_{i,j} c_i c_j d(x_i, x_j) \leq 0$ for all finite $\{x_i\}$ and all $\{c_i\}$ with $\sum_i c_i = 0$.

PROPOSITION 5 (When Eq. (3) is an honest distance). *d of Eq. (3) is of negative type if and only if the centred similarity matrix s is positive semidefinite, in which case $\sqrt{d}$ embeds isometrically into a Hilbert space and $d = \|\psi(\cdot) - \psi(\cdot)\|^2$ for the associated feature map ψ .*

Sketch. This is Schoenberg's theorem [4]: d is of negative type iff e^{-td} is a positive definite kernel for all $t > 0$ iff $d = \|\psi(x) - \psi(y)\|^2$ for a Hilbert map ψ ; expanding the square recovers Eq. (3) with $s(a, b) = \langle \psi(a), \psi(b) \rangle$, so negative type of d is the positive semidefiniteness of the centred s . See [9, 10] for the modern bi-Lipschitz refinements. $\square$ $\square$

ALIGNMENT SCORES NEED CORRECTION. General pairwise-alignment similarities (Smith–Waterman, BLAST) are *not* positive semidefinite, so Eq. (3) can violate $d \geq 0$ or the triangle inequality; canonicalising an alignment score into a metric requires an explicit correction [11]. In v3 this correction is the non-negativity clamp inside `seqtree`’s Gram transform — a projection of the score onto the nearest conditionally-negative-definite cone — which guarantees the metric axioms Prop. 1–2 rely on. The choice of s (BLOSUM62 here; `TCRBLOSUM` or a VDJ-specific matrix are alternatives) selects *which* metric; all are admissible provided the clamp restores negative type.

GAP-BLOCK VERSUS SMITH–WATERMAN. The v3 junction metric replaces a full Smith–Waterman optimisation by the best of four predetermined contiguous gap-block placements. When the optimal CDR3 alignment inserts a single contiguous gap inside the hypervariable core — the generic case once the conserved Cys/Phe anchors are fixed — the gap-block score equals the Smith–Waterman optimum; the two diverge only for multi-gap alignments of very unequal lengths. Empirically the induced coordinate systems are strongly but not perfectly aligned (Fig. 2: the same D – d correlation is $R = 0.64$ for Smith–Waterman and $R = 0.49$ for `gapblock`), so v3 embeddings are a *new, versioned* coordinate system rather than a bit-for-bit reproduction of the published one.

T.3. IMMUNE RECEPTORS AND THE RECOMBINATION MODEL

THE PROTOTYPE MEASURE IS P_{gen} . The diversity of an immune repertoire is astronomically large, but it is not arbitrary: it is the image of a low-entropy stochastic process, the V(D)J recombination model P_{gen} [2, 12, 3]. TCREMP takes the prototype-sampling measure of § T.1 to be exactly this model, so each coordinate $\varphi(x)_k = d(x, p_k)$, $p_k \sim P_{\text{gen}}$, measures the alignment distance from x to a typical product of recombination.

PROPOSITION 6 (Distance-profile functional). *For $p \sim P_{\text{gen}}$ the empirical distribution of $\{\varphi(x)_k\}_{k=1}^K$ converges weakly to the law of $d(x, p)$. Hence $\varphi(x)$ is a Monte-Carlo sketch of the map $x \mapsto \text{Law}_{p \sim P_{\text{gen}}} d(x, p)$, and two receptors with the same distance-profile-to- P_{gen} are asymptotically embedding-indistinguishable.*

Proof. The coordinates are i.i.d. evaluations of the fixed 1-Lipschitz map $p \mapsto d(x, p)$ under P_{gen} ; the Glivenko–Cantelli theorem gives uniform convergence of their empirical law to that of $d(x, p)$. $\square$ $\square$

This reframes what the embedding “learns”: not a language model of receptor sequences — which the entropy of P_{gen} (~ 40 bits for TRB) makes futile — but each receptor’s geometric relationship to the generative distribution itself. It also predicts that the embedding is insensitive to how the prototypes are obtained, provided they cover the same support.

PROPOSITION 7 (Prototype-source robustness). *Let μ, ν be two prototype measures that are mutually absolutely continuous with bounded density ratio $\rho = d\mu/d\nu$ on the support of the repertoire. Then the energy dissimilarities of Prop. 3 satisfy $\bar{D}_\mu^2(x, y) = \mathbb{E}_\nu[\rho(p)(d(x, p) - d(y, p))^2]$, so the two embeddings induce distances that agree up to the χ^2 discrepancy between μ and ν ; as $\chi^2(\mu \parallel \nu) \rightarrow 0$ the pairwise-distance vectors become perfectly correlated.*

Sketch. Importance sampling rewrites the μ -expectation as a ν -expectation with weight ρ ; a Cauchy–Schwarz bound on Cov under the two measures controls the correlation of the resulting distance vectors by $\chi^2(\mu \parallel \nu) = \mathbb{E}_\nu[\rho^2] - 1$. $\square$ $\square$

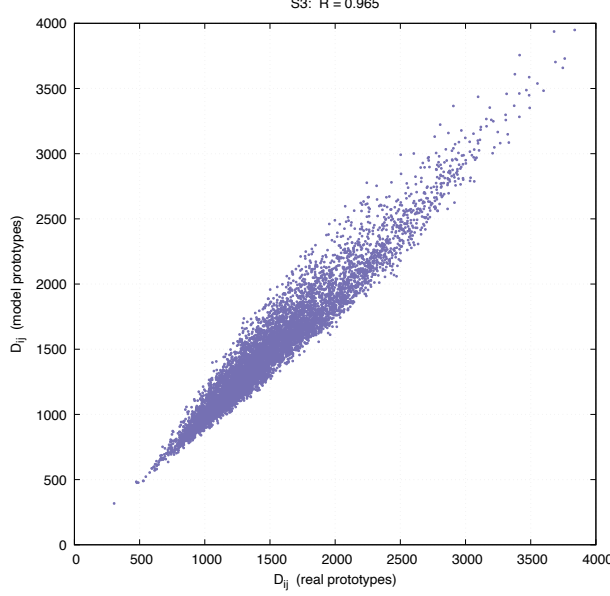


Figure 3: Pairwise embedding distances built from real (Britanova [13]) versus model (OLGA/Murugan [2, 3]) prototypes agree ($R = 0.97$; suppl. S3), confirming Prop. 7. `mir/bench/theory.py::prototype_source_correlation`.

A real healthy-donor repertoire (Britanova *et al.* [13]) and the synthetic recombination model (Murugan *et al.* [2]) are, after thymic selection, close in exactly this sense, which is why supplementary Fig. S3 finds their embedding distances correlated at $R = 0.96$ — reproduced here at $R = 0.97$ (Fig. 3).

DISTRIBUTION LAWS. The random construction also fixes the shape of the two histograms that dominate any embedding analysis.

PROPOSITION 8 (Dissimilarity and distance laws). *Under mild mixing of per-position alignment contributions: (i) the pairwise dissimilarity d_{ij} is asymptotically Gamma-distributed — a sum of many non-negative, bounded per-column penalties, so the central-limit tendency is tempered by positivity and right-skew; (ii) the embedding distance D_{ij} follows a generalized extreme value (Fréchet–Gumbel) law rather than a Gaussian, because $D = \|\varphi(x) - \varphi(y)\|_2$ is dominated by its largest coordinate differences, which are extreme-value statistics over the K prototypes.*

Sketch. (i) d is a sum of per-aligned-column Gram penalties (Eq. (3)), each in a bounded non-negative range; a Gamma law is the maximum-entropy fit to a positive sum with matched mean and variance, and dominates the Gaussian on the skew (Fig. 4, left). (ii) By Prop. 2 the ℓ^∞ part of D equals $d(x, y)$, an extreme over prototypes; the Fisher–Tippett–Gnedenko theorem [14] forces the maximum of K light-tailed coordinate differences into the GEV family, and the same tail governs ℓ^2 once distances concentrate in high dimension [15] (Fig. 4, right). □ □

Both laws are confirmed on v3 (Fig. 4): the Gamma fit to d_{ij} beats the Gaussian on AIC, and the GEV fit to D_{ij} beats the Gaussian decisively (Kolmogorov–Smirnov 0.021 versus 0.078). The

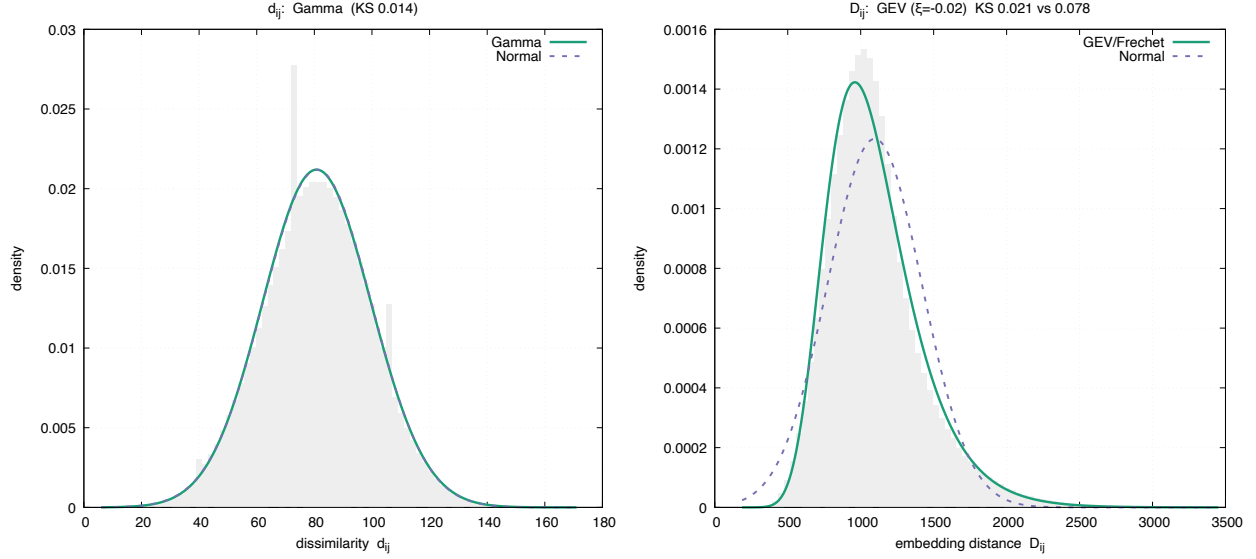


Figure 4: Distribution laws (Prop. 8; suppl. S1). Left: d_{ij} is Gamma, not Gaussian. Right: D_{ij} follows a GEV/Fréchet law (κ s 0.021) that the Gaussian (0.078) misses on the heavy right tail. `mir/bench/theory.py::fit_distributions`.

fitted shape $\xi \approx 0$ places v_3 at the Gumbel–Fréchet boundary, marginally lighter-tailed than the $\xi = +0.11$ of the published Smith–Waterman construction — a metric-dependent detail, not a qualitative difference.

T.4. TCRs, THREE-COMPONENT EMBEDDINGS, AND PAIRED CHAINS

THE PRODUCT METRIC. TCREMP emits three distances per prototype — a V-germline, a J-germline, and a junction distance (or CDR1, CDR2, junction) — interleaved into $\mathbb{R}^{3K}$ (`mir/embedding/tcremp.py`). Writing $\varphi = (\varphi_V, \varphi_J, \varphi_{\text{junc}})$ for the three blocks, the block structure of the Euclidean norm gives the exact decomposition

$$D(x, y)^2 = D_V(x, y)^2 + D_J(x, y)^2 + D_{\text{junc}}(x, y)^2, \quad (4)$$

so the embedding metric is the orthogonal sum of the per-component metrics.

PROPOSITION 9 (Paired-chain concatenation). *For a paired receptor (α, β) embedded by concatenating the per-chain maps, $D_{\text{pair}}(x, y)^2 = D_\alpha(x, y)^2 + D_\beta(x, y)^2$. More generally, concatenating landmark embeddings of independent factors realises the ℓ^2 product of the factor metrics.*

Proof. Concatenation stacks coordinate blocks; the squared ℓ^2 norm of a stacked vector is the sum of the squared block norms, which is Eq. (4) extended across chains. $\square$ $\square$

Paired $\alpha\beta$ TCR embedding (suppl. S4) and, prospectively, the concatenation of epitope- and MHC-groove coordinates into a single TCR–pMHC vector are therefore principled: each modality contributes an orthogonal block and distances add in quadrature.

THE GERMLINE BLOCKS ARE LOW-RANK. The V and J coordinates depend on the sequence only through its germline gene identity, which explains the redundancy that PCA removes.

PROPOSITION 10 (Germline rank bound). *Let G_V be the number of distinct V alleles present in a dataset S . The V-block matrix $\Phi_V \in \mathbb{R}^{|S| \times K}$ with rows $\varphi_V(x)$ has at most G_V distinct rows, hence $\text{rank } \Phi_V \leq G_V$. The same holds for J with G_J , and $G_V, G_J \ll K$.*

Proof. $\varphi_V(x)_k = d_V(v(x), v(p_k))$ depends on x only through its V allele $v(x)$ (the germline distances are a fixed allele $\times$ allele table, `mir/distances/germline.py`); receptors sharing a V allele have identical V-rows, so there are at most G_V distinct rows and the rank is bounded by G_V . $\square$ $\square$

A human TRB repertoire has $G_V \lesssim 60$, $G_J \lesssim 15$, against K in the thousands: two-thirds of the raw coordinates live in a subspace of dimension $\leq G_V + G_J$. The `StandardScaler`→PCA step (`mir/embedding/pca.py`) recovers this compact basis — Landmark MDS on φ — and an ANOVA of the leading components against gene usage recovers the germline axes, while the junction block carries the residual $\Theta(K)$ -dimensional specificity signal.

STRUCTURAL GROUNDING. That the metric is biologically meaningful, not merely combinatorial, is shown by supplementary Fig. S5: embedding distances between TCR:pMHC complexes correlate with the C α RMSD of their crystal structures. Formally this asserts a Lipschitz map from the sequence metric (X, d) into the structural metric, so that specificity groups — receptors binding one epitope — occupy small-diameter regions of embedding space, and clustering recovers them (Table S1; `mir/bench/metrics.py`).

T.5. IGH AND SOMATIC HYPERMUTATION

B-cell receptors differ from TCRs in one decisive respect: after recombination they undergo somatic hypermutation (SHM), a context-dependent point-mutation process [16]. Write M_m for an operator applying m substitutions to a germline-rearranged sequence.

PROPOSITION 11 (Bounded embedding drift). *There is a constant c (the maximal per-substitution Gram penalty) such that $d(x, M_m x) \leq c m$, and therefore $D(\varphi(x), \varphi(M_m x)) \leq \sqrt{K} c m$. Embedding drift under SHM is linear in the mutation load m .*

Proof. Each substitution changes at most one aligned column of Eq. (3), by at most the bounded penalty range c ; m substitutions change d by at most $c m$, and Prop. 1 propagates this to D with the $\sqrt{K}$ factor. $\square$ $\square$

Proposition 11 has two consequences. First, a clonal lineage under affinity maturation traces a path in embedding space whose step size is set by the per-generation mutation load, so the embedding preserves lineage proximity — the substrate for density and trajectory analysis (§T.6). Second, it explains why IGH is the hardest chain for the neural codec of Part 2 (forward cosine 0.79, exact reconstruction 0.16, against $\geq 0.93/0.50$ for TCR): (i) IGH CDR3s are long, so the fixed-length-40 gap-block tokenisation (`mir/ml/tokenize.py`) truncates an appreciable fraction of the variable middle, discarding information; (ii) SHM inflates the junction distance by $\Theta(m)$, so germline-rearrangement

prototypes sampled from P_{gen} systematically under-cover the mutated support (the density ratio ρ of Prop. 7 becomes large and the robustness guarantee degrades); and (iii) the per-receptor entropy is higher. The remedy the theory suggests is to separate the SHM-robust germline distance (V/J, Prop. 10) from the mutation-inflated junction distance and to augment the prototype set with a mutation model, i.e. to sample prototypes from P_{gen} pushed through M_m rather than from P_{gen} alone.

T.6. DENSITY GEOMETRY AND BACKGROUND SUBTRACTION

The embedding turns a repertoire into a point cloud in $\mathbb{R}^{3K}$ (or its PCA projection), and the natural null model is the pushforward of the generation measure.

DEFINITION 2. Let $f_{\text{gen}} = \varphi_{\#}P_{\text{gen}}$ be the density of embedded prototypes and f_{obs} the density of an observed repertoire on embedding space. The *enrichment* of a region is the density ratio $E(z) = f_{\text{obs}}(z)/f_{\text{gen}}(z)$.

PROPOSITION 12 (Continuous limit of neighbour enrichment). *The graph neighbourhood test of TCRNET/ALICE — comparing the observed count of receptors within an alignment radius r of z to its expectation under P_{gen} — converges, as $r \rightarrow 0$, to the density ratio $E(z)$. Convergent selection is precisely the set where $E(z) > 1$.*

Sketch. The observed and expected neighbour counts in a ball of radius r are $f_{\text{obs}}(z)$ and $f_{\text{gen}}(z)$ times the common ball volume $+o(r^d)$; their ratio tends to $E(z)$ as the volume shrinks, which is the standard consistency of a balloon density-ratio estimator. $\square$ $\square$

Proposition 12 recasts graph-based background subtraction as continuous density-ratio estimation on embedding space, which is the foundation for the TCRNET-without-graphs method of Part 2: estimate f_{gen} once from P_{gen} samples (or from the fast forward codec $\varphi \approx$ neural network, `mir/ml/encoder.py`), estimate f_{obs} by kernel density or a normalising flow, and report $E(z)$ directly — no neighbourhood graph, and a smooth enrichment field rather than per-vertex counts.

T.7. INSIGHTS, REPRODUCTION, AND OPEN PROBLEMS

A UNIFYING VIEW. The results above place TCREMP in a well-understood corner of metric geometry. The map φ is the empirical (Nyström) feature map of the alignment kernel; its ℓ^∞ read-out is the Kuratowski isometry, its ℓ^2 read-out is Landmark MDS, and its PCA is the classical scaling of the landmark columns. Three observations may be of independent interest beyond immunology: (a) taking the landmark measure to be a *generative model* of the data (P_{gen}) rather than a random subsample makes the embedding a sketch of each point’s relation to the data-generating process, and makes prototype choice provably robust (Prop. 7); (b) high-dimensional prototype-embedding distances obey a generalized extreme value law (Prop. 8), a clean consequence of distance concentration that is easy to mistake for Gaussian; and (c) the germline rank bound (Prop. 10) shows how categorical side-information injects exactly-low-rank structure into an otherwise full-rank distance embedding.

REPRODUCTION. Every quantitative statement is regenerated by `mir/bench/theory.py` and the data/figure scripts in this directory (`gen_theory_data.py`, the `*.gp` gnuplot scripts, `diagram_embedding.dot`); Table 1 collects the numbers.

Table 1: Theory predictions and their v3 reproduction (`mir/bench/theory.py`; $n = 2500$ CDR $_3\beta$). Paper values from [1], suppl. S1–S3.

Claim	Paper	mirpy (Smith–Waterman)	mirpy (gapblock v3)
S2 / Prop. 3: $\text{corr}(D_{ij}, d_{ij})$	0.56	0.64	0.49
S1 / Prop. 8: law of d_{ij}	Gamma $>$ Normal	Gamma $>$ Normal	Gamma $>$ Normal
S1 / Prop. 8: law of D_{ij}	GEV, $\xi = +0.11$	GEV $\gg$ Normal, $\xi \approx 0$	GEV $\gg$ Normal, $\xi \approx 0$
S3 / Prop. 7: real vs. model prototypes	0.96	—	0.97

OPEN PROBLEMS. Four questions are left open. (1) Sharp distortion constants for the lower bound complementary to Prop. 1 when prototypes are P_{gen} -sampled, i.e. a Bourgain-type [17] guarantee tuned to the recombination measure rather than to the worst case. (2) The exact extreme-value index ξ of Prop. 8 as a function of the substitution matrix and prototype count. (3) A mutation-corrected metric for IGH that makes Prop. 7 hold under SHM (§T.5). (4) Consistency and rates for the density-ratio estimator of Prop. 12 on the PCA-projected embedding, which would put continuous background subtraction on the same footing as the graph tests it replaces.

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